Autocorrelation Metrics to Estimate Soil Moisture Persistence From Satellite Time Series: Application to Semiarid Regions

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***Abstract*— Satellite-derived soil moisture (SM) products have become an important information source for the study of land surface processes in hydrology and land monitoring. Characterizing and estimating soil memory and persistence from satellite observations is of paramount relevance, and has deep implications in ecology, water management, and climate modeling. In this work, we address the problem of SM persistence estimation from microwave sensors using several autocorrelation metrics that, unlike traditional approaches, build on accurate estimates of the autocorrelation function from nonuniformly sampled time series. We show how the choice of the autocorrelation estimator can have a dramatic impact on the SM persistence metrics derived thereof, particularly given the nonuniform nature of satellite observations, yet this fact has been overlooked to a large extent in literature. We give empirical evidence of performance using ground-based SM measurements, as well as L-band Soil Moisture and Ocean Salinity (SMOS) and C-band [Advanced Microwave Scanning Radiometer-2 (AMSR2), Advanced Scatterometer (ASCAT)] remotely sensed SM data. Experiments along transects allow us to scrutinize the inter-method consistency and the spatial–temporal characteristics of autocorrelation estimators. This motivates the introduction of novel measures of spatial–temporal autocorrelation and allows us to retrieve improved persistence estimates. Results over the Iberian Peninsula indicate the SM persistence patterns captured by L- and C-band microwave sensors over semiarid regions exhibit spatially concurrent patterns of persistence and support their combination in long-term data records. We conclude that accounting for the nonuniform nature of the satellite time series using robust autocorrelation estimations allows providing improved measures and spatial descriptions of SM persistence.**

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***Index Terms*— Advanced Microwave Scanning Radiometer-2 (AMSR2), Advanced Scatterometer (ASCAT), autocorrelation, e-folding time, Least Absolute Shrinkage and Selection Operator (LASSO), Lomb-Scargle periodogram, microwave sensors, persistence, soil moisture, Soil Moisture and Ocean Salinity (SMOS), spatial–temporal.**

I. Introduction

SOIL Moisture is an essential climate variable (ECV) that regulates the exchange of water and heat energy between the land and the atmosphere [1]. Due to its memory characteristics, it is an important contributor to climate persistence on land, impacting the development and duration of natural hazards such as droughts, floods, and heatwaves [2]–[4]. Knowledge of the timescales of soil water variability and storage are of significance to a variety of disciplines, such as agriculture [5], ecology [6], and climate [7]–[9]. The estimation of these timescales from Earth observation-based soil moisture (SM) products is an area of intense current research [10]–[13].

The autocorrelation function has been widely used for estimating persistence in time series of atmospheric, land, or oceanic Earth system variables, for diverse application settings [14]. Here, we characterize the persistence or "memory" of different satellite SM time series by means of autocorrelation-based metrics. These metrics are in general a powerful tool to confront ground-based, observed, and modeled SM products, since they measure the SM response to hydrological processes that, while occurring at different spatial scales, lead to a common temporal signature [15]–[17]. In particular, they allow investigating the internal structure and parameterizations used in land surface models to represent SM inertia and surface water loss mechanisms, i.e., evapotranspiration, infiltration, and drainage [12], [18]. Given the large uncertainties associated with interactions of these nonlinear processes [19], [20], measures of temporal autocorrelation from satellite SM products could guide the parameter optimization of Earth system models. Also, a main motivation for investigating SM memory characteristics is the possible use of SM initialization for seasonal forecasting [21]. The potential of SM for seasonal forecasting is likely

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to be highest in regions where both SM memory and land–atmosphere coupling are important.

At present, the applicability of autocorrelation-based metrics in observational time series faces two major obstacles: 1) robust computation is needed in the presence of missing data and/or unevenly sampled time series; and 2) stable estimates are required to summarize the information conveyed by the autocorrelation function in a single expressive parameter or indicator. Previous studies that estimate SM persistence from satellite data have not fully addressed the importance of choosing a robust autocorrelation estimator that accounts for their nonuniform nature. In this work, we propose two nonparametric estimators of the spectrum for nonuniformly sampled time series, in such a way that the regular frequency grid in the spectrum allows us to provide an estimation of the autocorrelation sequence and from there obtain improved spatial descriptions of SM persistence from satellites. The proposed estimation algorithms, based on the Lomb-Scargle (LS) periodogram [22] and on the Least Absolute Shrinkage and Selection Operator (LASSO) [23], use an inverse Fourier transform of the estimated spectrum for yielding the uniformly sampled autocorrelation. This allows us to compare the autocorrelation function with respect to other existing estimators, and to scrutinize the quality of the autocorrelation-based metric obtained to characterize the persistence of the time series.

We use the proposed metrics to investigate the surface SM memory captured by three spaceborne microwave sensors, namely, the European Space Agency (ESA's) Soil Moisture and Ocean Salinity (SMOS) L-band radiometer, the Japan Aerospace Exploration Agency's (JAXA's) C-band Advanced Microwave Scanning Radiometer-2 (AMSR2), and the Eumetsat C-band Advanced Scatterometer (ASCAT). The three sensors are integrated in the Copernicus Climate Change Service (C3S) long-term SM product [24], [25], as well as in the National Oceanic and Atmospheric Administration (NOAA) Satellite Soil Moisture Operational Product System (SMOPS) [26] and other recent initiatives to merge multisatellite satellite SM products [27]. Within the electromagnetic spectrum, low-frequency microwaves in the range of 1–10 GHz are most useful for sensing SM. Within this range, lower frequencies (L-band) can pass further through the canopy and soil, and are more sensitive to SM. In comparison to L-band, C-band frequencies have a shallower soil sensing depth (top 1 cm versus 5 cm) and a larger attenuation in presence of vegetation [28]. Microwave instruments can either be passive (radiometer) or active (radar). Radiometers observe the radiation naturally emitted from the land surface, and typically have a coarse resolution ($\sim$25 km). Radars transmit an electromagnetic wave and measure the amount of backscatter. Spaceborne real aperture radars (scatterometers) provide spatial resolutions comparable to those from radiometers, while synthetic aperture radars can yield dramatically higher resolutions (meter-scale). In the present study, we analyze the temporal autocorrelation of three coarse resolution microwave-based SM products obtained from three distinct sensors: an L-band radiometer (SMOS), a C-band radiometer (AMSR2), and a C-band scatterometer (ASCAT). Although satellite microwave data are not affected by clouds, measurements can be nonetheless limited in some regions due to the presence of radio frequency interferences, presence of seasonal snow and/or freezing temperature masking observations, or high uncertainty in their inversion algorithm (e.g., dense vegetation, high topography) [28]. Hence, dealing with nonuniformly sampled time series is a necessary step to compute robust estimates of the autocorrelation function.

In this article, we propose novel measures of spatial–temporal autocorrelation that allow us to characterize the persistence of SM anomalies in time and space using relatively short (4 years) microwave satellite time series. Our study focuses on the Iberian Peninsula, which provides an ideal controlled setting for the study of satellite-based SM dynamics; its climate is semiarid and hence with a water limited dominant evaporation regime, SM displays a large annual cycle and it is not densely vegetated [29]. Also, as spring-time SM anomalies over the Iberian Peninsula have been shown to be a precursor to droughts and heatwaves in Europe [30], [31], the SM memory characterization in this region is of particular interest for further investigation of this phenomena. We evaluate to what extent the proposed measures of persistence can be impacted by the time series length as well as by missing data due to the presence of seasonal snow. Furthermore, we explore whether different sensor characteristics such as the operating frequency (C- versus L-band), the sensor type (active versus passive), and the temporal frequency of the observations have an impact on the estimated SM persistence patterns. To do so, the autocorrelation function and derived persistence measures obtained from SMOS, AMSR2, and ASCAT satellite data are first compared to the ones obtained with collocated *in situ* SM measurements from the REMEDHUS network (central part of the Duero basin). Then, the spatial patterns of persistence obtained from the three satellites across the Iberian Peninsula are analyzed to identify common features and characterize interdependencies with SM regimes. We conclude that the proposed autocorrelation estimators allow comparing the temporal dynamics captured by different sensors and platforms and provide consistent spatial descriptions of SM persistence from observational data.

The remainder of this article is organized as follows. In Section II, we fix notation and review the different autocorrelation estimators for nonuniform time series. In Section III we first introduce the physical models describing SM temporal variability and their approximations to estimate persistence using autocorrelation properties. Subsequently, we introduce the SM data sets and their effective temporal sampling over our region of study. Section IV shows the results of the proposed autocorrelation estimators in different experimental set-ups. First, the autocorrelation curves of satellite and *in situ* data are compared at the REMEDHUS network. Second, we analyze the impact on the estimators of the time series length and of missing data during winter. We then focus on the spatial–temporal characteristics of the autocorrelation estimates and summarize their information into two single expressive parameters, namely, the e-folding time (τ_e), and the value of the autocorrelation for a time delay (intermediate between zero lag and the first zero crossing). These parameters are subsequently used to investigate and confront the persistence patterns captured by

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the three satellite products and their relation with SM regimes. Conclusions and perspectives from this work are given in Section V.

II. Autocorrelation From Nonuniform Series

A. Signal Notation and Spatial–Temporal Autocorrelation

SM evolution can be seen as a stochastic and spatially distributed time process, for which we propose the following simple signal model. Let $U(t, \mathbf{s})$ be a stochastic process defining the spatio-temporal evolution of SM, where t denotes continuous time and $\mathbf{s}$ denotes the continuous-surface space vector coordinates, and let $u(t, \mathbf{s})$ denote the observable fluctuations from this evolution, obtained by sampling at a set of time instants t_j, this is

$$u(t, \mathbf{s}) = \sum_{j=1}^{N} U(t, \mathbf{s}) \delta\big(t - t_j\big) \tag{1}$$

where N is the total number of observed samples through a given measurement window of duration T. For instance, measurements can be taken within 1-day sampling intervals, but some of the measurements could be missing or not fully reliable for some days because of heavy precipitation or acquisition issues.

Let us assume that the measurement can be cast as a mathematical complex transformation Γ of the SM parameter

$$m(t, \mathbf{s}) = \Gamma(u(t, \mathbf{s})|\ell) \tag{2}$$

where ℓ denotes the sensor that is taking the measurement. In this work, ℓ corresponds to either SMOS, AMSR2 or ASCAT sensors. It is often convenient to represent in a clear way the spatial and temporal dynamics of satellite measurements on SM, for which we use the previously proposed M-mode representation (e.g., [32]). The M-mode of the space-multidimensional measurement function as its temporal evolution is often known as a transect representation in Earth Sciences [33]. A longitudinal (or latitudinal) transect passing through location $\mathbf{s}_0$ is here used and denoted by $\mathbf{s}_0^l$, so that the M-mode representation is given by $m(t, \mathbf{s}_0^l)$, where l is a conveniently selected longitudinal (latitudinal) line.

In the long term, time stationarity can be assumed for SM at each given location. Then, the autocorrelation for a specific location $\mathbf{s}_0$ is given by [34]

$$R_m(\tau, \mathbf{s}_0) = \mathbb{E}_t\{m(t + \tau, \mathbf{s}_0) m(t, \mathbf{s}_0)\}. \tag{3}$$

The M-mode of the autocorrelation for a line l through location $\mathbf{s}_0$ is denoted by $R_m(\tau, \mathbf{s}_0^l)$, and called an autocorrelation transect for short in the rest of this article. In addition, a well-known relationship [14] is that the Power Density Spectrum (PSD) at a given location $\mathbf{s}_0$ is related with the autocorrelation through the Fourier Transform, this is

$$S_m(f, \mathbf{s}_0) = \mathcal{F}\{R_m(\tau, \mathbf{s}_0)\} \tag{4}$$

where $\mathcal{F}$ denotes the Fourier Transform, $S_m(f, \mathbf{s}_0)$ denotes the temporal PSD of the SM stochastic process at location $\mathbf{s}_0$, and f denotes temporal frequency.

We can now think of using some method for estimating the spectrum from the observed samples, that is

$$\widehat{S}^{\nu}(f, \mathbf{s}_0) = \Theta(m(t, \mathbf{s})|\nu) \tag{5}$$

where Θ denotes the procedure operator for estimating the spectrum from the nonuniformly time-distributed measured samples, following some estimation criterion ν, and $\widehat{S}^{\nu}(f, \mathbf{s}_0)$ is the spectral estimation given by said criterion. The estimated time autocorrelation for that location is then

$$\widehat{R}_m^{\nu}(\tau, \mathbf{s}_0) = \mathcal{F}^{-1}\{\widehat{S}^{\nu}(f, \mathbf{s}_0)\} \tag{6}$$

where $\mathcal{F}^{-1}$ denotes the inverse Fourier transform. Note that the estimated autocorrelation will depend both on sensor ℓ and on estimation method ν. In this work, ν is scrutinized for several estimation criteria, namely, the *naïve*, LS and LASSO estimators, that are described next, and the first-order autoregressive AR(1) estimator, that will be introduced in Section III-A.

B. Estimation Algorithms

In order to estimate the spectrum in (5), we first fix a matrix notation for the observed samples. The time instants are grouped in a sampling-times vector

$$\mathbf{t} = [t_1, t_2, \ldots, t_N] \tag{7}$$

and the corresponding observed samples from a satellite measurement system in vector notation for that location can be expressed as

$$\mathbf{m}(\mathbf{s}_0) = [m(t_1, \mathbf{s}_0), m(t_2, \mathbf{s}_0), \ldots, m(t_N, \mathbf{s}_0)] \tag{8}$$

which, for short of notation, we will denote as $\mathbf{m}$ hereafter.

A simple autocorrelation estimator for the case when time instants correspond to uniform sampling with Δt sampling period is simply given by

$$\widehat{\mathbf{R}} = \mathbf{m} * (\mathbf{m}\mathbf{J}) \tag{9}$$

where $\widehat{\mathbf{R}}$ denotes the vector with the autocorrelation estimated samples, $\mathbf{J}$ is the anti-diagonal identity matrix with size $N \times N$ yielding the flipped vector for $\mathbf{m}$, and $*$ denotes the vector convolution. Note that this vector yields the estimated samples for $\widehat{R}_m(\tau, \mathbf{s}_0)$ with lag $\Delta\tau = \Delta t$, with $\tau = 0$ in the central sample of this vector. We call this approach the *naïve* estimator, and its application should work well with uniformly sampled time series, but it is expected to degrade with the increasing number of missing samples in the time series.

More advanced estimators can be obtained by following the mixed sinusoidal signal model as described hereafter. Let us assume a frequency resolution Δf and a maximum frequency f_M, in order to build a frequency grid, denoted in vectors as

$$\mathbf{f} = [\Delta f, 2\Delta f, 3\Delta f, \ldots, f_M] \tag{10}$$

which has up to $N_f = (f_M/\Delta f) + 1$ elements. We assume that the observation vector can be approximated by a signal model consisting of the summation of sinusoidal signals on each frequency in the grid, this is

$$\widehat{\mathbf{m}} = \sum_{i=1}^{N_f} a_i \cos(2\pi i \Delta f \ \mathbf{t} + \phi_i) \tag{11}$$

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which can be readily linearized with simple manipulations, and expressed in matrix form as follows:

$$\widehat{\mathbf{m}} = \sum_{i=1}^{N_f} c_i \cos(2\pi i \Delta f \mathbf{t}) + d_i \sin(2\pi i \Delta f \mathbf{t}) = \mathbf{z}\mathbf{X}^\top \quad (12)$$

where $c_i = a_i \cos(\phi_i)$ and $d_i = -a_i \sin(\phi_i)$; Vector elements are $\mathbf{c} = [c_1, \ldots, c_{N_f}]$ and $\mathbf{d} = [d_1, \ldots, d_{N_f}]$; Defining matrices $\mathbf{C} := \{\cos(2\pi i \Delta f t_j); i = 1, \ldots, N_f; j = 1, \ldots, N\}$ and $\mathbf{D} := \{\sin(2\pi i \Delta f t_j); i = 1, \ldots, N_f; j = 1, \ldots, N\}$, data matrix $\mathbf{X}$ is $\mathbf{X} = [\mathbf{C}, \mathbf{D}] \in \mathbb{R}^{N \times 2N_f}$; And phase and quadrature unknown coefficients are assembled in a single unknown vector $\mathbf{z} = [\mathbf{c}, \mathbf{d}] \in \mathbb{R}^{1 \times 2N_f}$.

1) Least Squares Regression: An improved estimation criteria needs to account for the presence of nonuniform sampling, and we propose two simple approaches based on linear regression using different minimization functionals. First, we propose the minimization of the L_2 norm of the residuals, given by

$$\widehat{\mathbf{z}} = \underset{\mathbf{z}}{\operatorname{argmin}} \|\mathbf{e}\|^2 = \underset{\mathbf{z}}{\operatorname{argmin}} \|\widehat{\mathbf{m}} - \mathbf{z}\mathbf{X}^\top\|^2 \quad (13)$$

which is given by Wiener–Hopf normal equation that reduces to compute the pseudo-inverse of $\mathbf{X}$ as $\widehat{\mathbf{z}} = \widehat{\mathbf{m}}\ \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}$.

This represents a regression data model of the observed data on the nonuniformly sampled sinusoids, and it is often called Least Squares Spectral Analysis (LSSA) [35]. Some modifications to this basic formulation were proposed by Lomb and Scargle [22], using a time delay such that sines and cosines at a given frequency in the grid are mutually orthogonal at sampling time instants, to obtain a better estimate of the power in that frequency. The estimated and so-called LS spectrum at the grid frequencies is given by

$$\mathbf{s}_+ = \frac{1}{\sigma_m^2}\left(\hat{\mathbf{c}}^{\odot 2} + \hat{\mathbf{d}}^{\odot 2}\right) \quad (14)$$

where $\mathbf{s}_+$ is the positive part of the estimated spectrum $\hat{\mathbf{z}}$ from (13), $\hat{\mathbf{c}}$ and $\hat{\mathbf{d}}$ are its subvectors, $^{\odot 2}$ denotes the element-wise power in a vector, and σ_m^2 denotes the variance of observed vector $\widehat{\mathbf{m}}$.

The complete spectrum is given by the following symmetric spectral vector:

$$\widehat{\mathbf{s}} = [\mathbf{s}_+\mathbf{J}, 0, \mathbf{s}] \in \mathbb{R}^{1 \times 2N_f + 1} \quad (15)$$

and then the estimated autocorrelation is

$$\widehat{\mathbf{s}} = \widehat{\mathbf{s}}\ \mathbf{F}^{-1} \quad (16)$$

where $\mathbf{F}$ denotes the Fourier transform matrix for vectors in $\mathbb{R}^{1 \times 2N_f + 1}$.

2) Sparse-Promoting Regression: As a second approach we considered the LASSO estimation method. This regression analysis was initially proposed in geophysics and has been later extensively used in a variety of problems in statistics and machine learning [23], [36]. It consists on regularizing the LS solution with the L_1 norm of the estimated vector, which has an effect of projecting to zero some small-amplitude elements of the solution and thus promoting sparse solutions. The LASSO estimator for our sinusoid-regression problem is given by

$$\widehat{\mathbf{z}} = \underset{\mathbf{z}}{\operatorname{argmin}} \|\widehat{\mathbf{m}} - \mathbf{z}\mathbf{X}^\top\|^2 + \eta\|\mathbf{z}\|_1 \quad (17)$$

where η is a regularization parameter, typically tuned by cross-validation. Once solved for $\widehat{\mathbf{z}}$, the autocorrelation estimation steps are similar to (14)–(16).

C. Toy Example on Synthetic Data

We performed a simple experiment on synthetic data to illustrate the use of the presented estimation methods. First, we created a 1000-samples signal by summing a sine and cosine waves of amplitudes 1 and 3, and at frequencies $f_1 = 0.5$ Hz and $f_2 = 3$ Hz, respectively. Nonuniform sampling was implemented by considering that only 150 samples were available during 6 s, and the rest are missing values. The actual and the estimated spectra of the signal when using LS and LASSO methods, as well as the corresponding autocorrelation functions are compared in Fig. 1. It can be seen that the LS estimation exhibits spurious peaks in its spectrum, and the main lobe of the estimated autocorrelation tends to be narrower than the actual one. The LASSO estimation, in turn, does not exhibit spurious peaks and seems to provide a better estimation of the autocorrelation main lobe. For increasing lags, there is a tendency of LS to provide lower autocorrelations than the actual one and of LASSO to provide higher autocorrelations than the actual one. This simple experiment reveals that the choice of the autocorrelation estimation method can highly impact the estimated parameters characterizing the SM memory.

D. Remarks and Practical Considerations

Several practical considerations need to be taken into account when working with irregularly sampled time series [22]. The first consideration refers to the high-frequency limit f_M in the case of nonuniform sampling. An adequate maximum frequency is necessary to avoid the loss of relevant information, and some proposals for a Nyquist-like frequency in the case of nonuniform sampling can be found in the literature. Some studies suggested to use the mean [37], [38] or more recently the median [39] of the sampling intervals. However, none of them gave good quality spectral estimators in our problem, since we need to keep high-frequency components, which are larger than the limits imposed by those approaches. For the low-frequency limit, a value of $1/T$, with T the time length of the samples, or just a zero value can be considered, because it is unlikely to add any significant spurious peak to the spectral estimators. A well-motivated high-frequency limit can be obtained from the prior knowledge of the signals under analysis [22].

The second consideration refers to the frequency range. It is important to determine a frequency-grid spacing smaller than the expected widths of the periodogram peaks. An inadequate grid can miss narrow yet important peaks in the spectrum representation of our signal. The resolution of the frequency axis can be defined by using an oversampling parameter δ_f.

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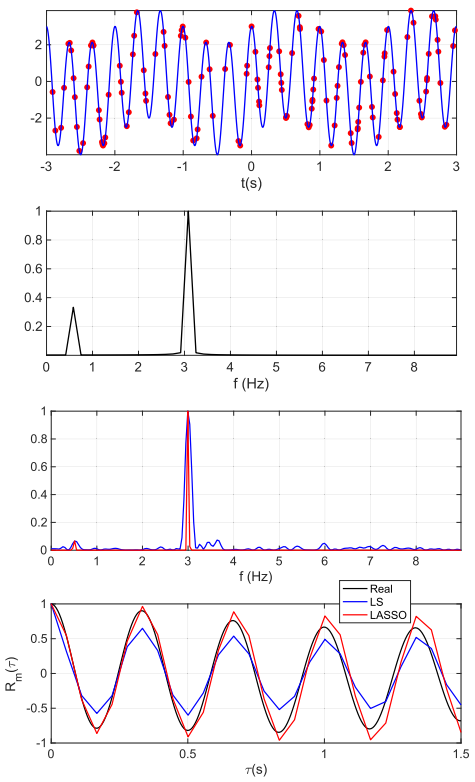


Fig. 1. Synthetic data toy example with nonuniform sampling. The first panel shows the sampled signal (red points) and the regularly sampled version (blue), and the second their actual sinusoidal spectrum. The two following panels show the estimated spectra for LS and with LASSO, normalized to unity amplitude for visualization purposes, and the actual and estimated autocorrelations.

Thus, the size grid will be $\Delta f = 1/(\delta_f \times T)$, and therefore, $N_{\text{eval}} = f_{\max}/\Delta f$ periodogram evaluations will be required. Although the value of this parameter depends on the characteristics of the data under study, values ranging from $\delta_f = 5$ to 10 seem to work fine in the literature [22]. Note that an appropriate selection of the frequency-grid spacing will also avoid unnecessary evaluations of the spectral estimator, and that it is recommended to adjust the limiting frequency to a minimum, based on prior information about the data.

III. Experimental Setup

A. Modeling SM Persistence

The SM memory is a measure of the time that a moisture anomaly is detectable and during which it can influence the atmosphere. Land–atmosphere coupling and soil memory concepts are interconnected, and are a result of complex and nonlinear interactions between multiple processes within the climate system [8], [9]. The physics behind SM memory can be examined with the standard surface water budget equation

$$\Delta z \frac{d\theta}{dt} = P(t) - \text{ET}(\theta, t) - D(\theta, t) - Q(\theta, t) \tag{18}$$

where θ is volumetric SM (–), t is time (days), Δz is the depth of the soil volume below the surface (mm), $P(t)$ is the precipitation rate (mm·d^{-1}), and water losses from the volume include evapotranspiration ET(θ, t), drainage $D(\theta, t)$, and runoff $Q(\theta, t)$ (mm·d^{-1}). The relative dominance of the water loss fluxes depends on how much time has passed since rainfall [40]. Runoff, when present, ceases only minutes after rainfall. Drainage occurs on a timescale of hours. In this work, we use satellite sensors with a temporal resolution on the order of 1–3 days (see Table I) and thus both can be assumed as negligible [12]. This results in water losses dominated by stage two ET (water limited conditions), where ET is linearly related to available soil water, and the resulting surface water equation can be modeled as a first-order Markov process under the assumption of white-noise rainfall forcing [15]. This approximation is widely adopted to approximate SM dynamics in climate systems [11], [16].

The temporal variability of SM, when modeled as a red-noise or first-order Markov process, is expressed as

$$\Delta z \frac{d\theta}{dt} = P(t) - \lambda\theta \tag{19}$$

where λ is a decay term that represents a linear dependence of ET on θ under water limited conditions, and $P(t)$ is assumed to be a white noise process. Physically, it represents a stochastic (noisy) process possessing some memory and an inherent exponential damping, which is continually forced by a random process. Note that for (19) to describe weather/climate fluctuations, the seasonal cycle has to be removed from the SM time series. In this modeling framework, time series of SM anomalies (deviations from the mean state) follow an exponentially decaying autocorrelation function given by:

$$r(k) = \exp(-k\lambda) \tag{20}$$

where k is the lag and $\tau_e = 1/\lambda$ is the decorrelation or e-folding time scale, the time lag at which the autocorrelation decreases to a factor of $e^{-1} \approx 0.37$. The e-folding time is typically used as a time scale of SM memory [17]. Other memory metrics based on the autocorrelation function have also been used in the literature, including the integral time scale [11], the autocorrelation for a fixed time lag [8], [21], or the SM variance spectrum [41]. However, as discussed in Section I, their applicability to recent satellite SM data records remains challenging. Also, note that many previous studies estimated SM memory from approximately monthly observations, thus missing substantial short-term variability. As noted in [10], this missing variability could lead to overestimation of the memory timescale.

More generally, the approximation of SM fluctuations as a red noise process in (19) can be described with a first-order autoregressive model AR(1), where a fluctuation depends only on its own immediate past plus a random component. This is a simple yet widely used approach for describing the autocorrelation structure in many climatic processes [42], [43]. The AR(1) model for uneven spacing can be numerically adjusted to estimate τ_e [44] or the lag-one autocorrelation coefficient [45] as measures of persistence. This approach has been recently applied to estimate a global map of the AR(1) coefficient using 3 years of satellite-based SM estimates [45]. In Section IV, we report important differences between persistence estimates obtained from fitting the AR(1) model and those from approximating the autocorrelation function. This

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is actually grounded on fundamentals of signal processing, and in particular on the challenging problem of estimating the autocorrelation function from unevenly sampled time series.

B. Data Sets

All data sets used in this study are listed in Table I. The period of study is four years, starting in July 2012. All satellite products were temporally composited at the daily scale and spatially aggregated to a common grid (the EASE2 25-km) using standard bilinear interpolation.

Hourly SM measurements acquired at 18 REMEDHUS stations were first averaged into daily values and then spatially averaged to obtain the network average. Each station within the network is equipped with capacitance probes (Hydra Probes of Stevens Water Monitoring System, Inc.) installed horizontally at a depth of 5 cm, with a reported accuracy of 0.003 m^3 m^{-3}. Previous studies have shown the stability of daily SM measurements in REMEDHUS and the convenience of using the daily network average for satellite validation [46], [47].

A standard requirement in remote sensing time series analysis is that of discounting the strong seasonal and interannual variability. The studies presented in this article rely on time series autocorrelation, which can be strongly influenced by deterministic low-frequency variability. Hence, prior to analysis, we removed the effect of seasonality on SM time series following the procedure in [45], which is well suited to calculate anomalies from relatively short time series. It consists on: 1) calculating the daily average values of the variable over all observed years; 2) concatenating three copies of the annual climatology as a single time series to ensure exact periodicity at its extremes; 3) smoothing the time series with a 61-day box filter chosen to minimize the effects of synoptic variability; and 4) subtracting the central year of the smoothed climatology from the original data to obtain daily anomaly series. Hence, we focus on characterizing how SM anomalies persist in time and space.

C. Temporal Frequency Considerations

SM memory estimates are dependent on the temporal sampling frequency of the observations. In this work, we use three satellites with sun-synchronous orbits, whose products are typically derived separately for ascending and descending overpasses with different local acquisition times.

For SMOS, morning passes (6 AM) are generally preferred to afternoon passes (6 PM) to ensure near-surface isothermal conditions and minimize Faraday rotation effects. Nonetheless, differences are minor, and the combination of measurements from the two orbits has been adopted in a variety of studies to maximize data coverage (e.g., [50], [51]). Likewise, we averaged SMOS ascending and descending orbits into daily estimates and show results for the SMOS combined product only. The ESA Climate Change Initiative (CCI)-PASSIVE product includes AMSR2 observations acquired during descending (1.30 PM) orbits only. This is consistent with the impact of non-near-surface isothermal land surface temperature conditions at 1.30 AM in the retrieval algorithm [24], [49], [52]. The ESA CCI-ACTIVE product includes

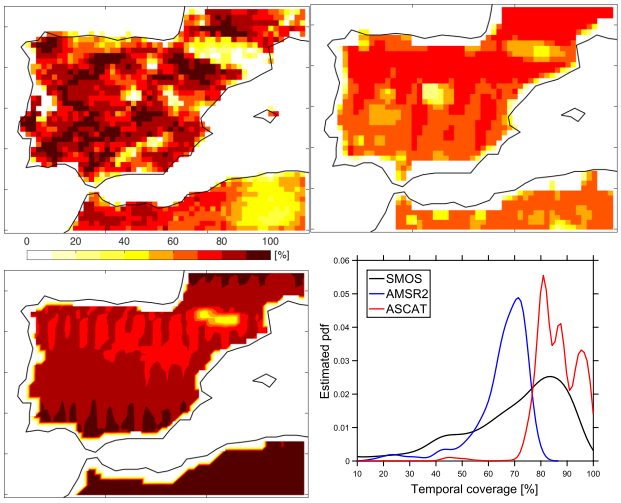


Fig. 2. Spatio-temporal data coverage of SMOS (top-left), AMSR2 (top-right) and ASCAT (bottom-left) SM satellite maps over the Iberian Peninsula for the study period (2012–2016), and comparison of the density of observations per pixel and satellite product.

ASCAT observations acquired during ascending (9.30 AM) and descending (9.30 PM) orbits, since the retrieval approach is based on change detection and therefore avoids the influence of diurnal variations in land surface temperature impacting on the uncertainty of the estimates [53], [54]. See [55] for a more detailed discussion on the impact of local acquisition times in microwave satellite SM retrievals.

The temporal coverage of the satellite products over REMEDHUS at the daily scale is 93.28% for SMOS, 73.27% for AMSR2, and 80.13% for ASCAT. The spatio-temporal coverage of the three satellites across the Iberian Peninsula (40.0–44.8° N, 11.0° W–4.8 N) is shown in Fig. 2. Overall, it can be seen that ASCAT provides the highest temporal coverage, of more than 80%, and AMSR2 the lowest, about 70% on average. SMOS temporal coverage is the most heterogeneous, with most pixels having about 80% coverage but with a heavy-tailed distribution toward lower values.

IV. Results

This section starts by studying the correspondence of the autocorrelation function estimates using ground-based SM measurements from the REMEDHUS catchment and collocated satellite-based SM estimates. Then we analyze the robustness to the severity of nonuniform sampling. Subsequently, a dedicated analysis to study the effect of seasonality and of the time series length on the autocorrelation estimators' consistency is undertaken. The quality of autocorrelation estimation in nonuniform sampling conditions as acquired from satellites is analyzed, where we show that simple spatial–temporal processing after (16) allows to enhance its signal-to-noise ratio, and we study the impact of noise cancellation on the parameter maps characterizing SM persistence. Finally, the best autocorrelation estimators and derived parameters are applied to investigate and confront the SM persistence patterns captured by the three satellite products over the Iberian Peninsula.

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TABLE I

DATA SETS OF SM USED IN THIS STUDY. THE REPRESENTATIVE SOIL LAYER DEPTH OF MICROWAVE SATELLITES IS AN AVERAGED THEORETICAL VALUE. NOTE THAT IT IS FREQUENCY-DEPENDENT BUT ALSO IT RAPIDLY DECREASES WITH INCREASING SOIL WETNESS [28]

Dataset	Version	soil layer depth	Spatial support	Temporal res.	Derived from	Reference
SMOS (BEC L3)	2.0	top 5 cm	Global EASE2 25 km	3 days	Satellite passive L-band	[49]
AMSR2 (ESA CCI PASSIVE)	3.3	top 1 cm	Global 0.25°	near daily	Satellite passive C-band	[24], [50]
ASCAT (ESA CCI ACTIVE)	3.2	top 1 cm	Global 0.25°	near daily	Satellite active C-band	[24], [50]
REMEDHUS	-	5 cm	Area of 35 x 35 km	hourly	18 ground-based stations	[47]

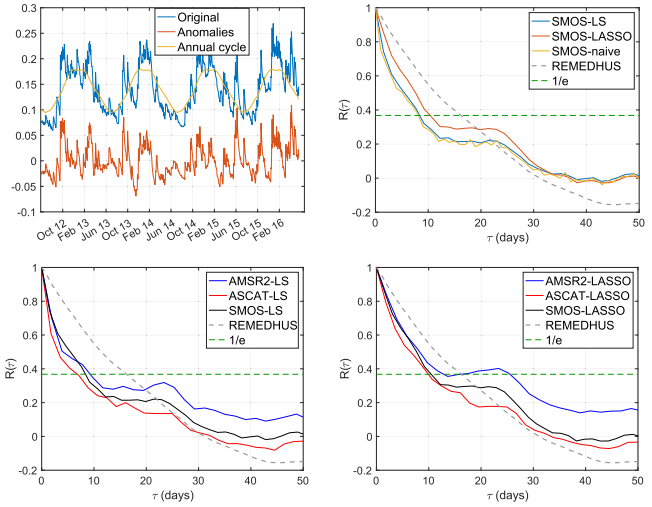


Fig. 3. Comparison between the autocorrelation functions of REMEDHUS *in situ* SM (m^3m^{-3}) measurements and collocated SMOS, AMSR2 and ASCAT SM time series. Illustration of *in situ* original time series (average of 18 stations), and anomalies after deseasoning (top-left). The autocorrelation of *in situ* SM anomalies is shown as gold standard (dashed gray) alongside: the estimated autocorrelation functions for the SMOS SM time series with LS, LASSO, and *naïve* methods (top-right), the estimated autocorrelation functions for the three satellite data sets using the proposed LS (bottom-left) and LASSO (bottom-right) algorithms. The value $R = 1/e$ is depicted as an horizontal dotted line.

A. REMEDHUS Ground Versus Satellite Autocorrelation

The REMEDHUS network is located at the central part of the Duero river basin in Spain. It covers a semiarid continental-Mediterranean agricultural region of 35 km × 35 km (41.1–41.5° N, 5.1–5.7° W) of approximately 1300 km^2. The homogeneity of the area makes it ideal for satellite product validation, as shown by the number of validation exercises conducted in the network (e.g., [29], [46], [49], [56]). In this work, ground-based REMEDHUS SM time series (average of 18 stations) are compared to collocated satellite SM estimates.

The REMEDHUS SM ground-based data were used as the gold standard to evaluate the temporal dynamics sensed by the three satellites at this location, cf. Table I. Fig. 3 (top-left) illustrates the *in situ* time series evolution (average of 18 stations) for the 4-year study period, its estimated seasonality or annual cycle used as typical behavior, and the anomaly time series obtained after its subtraction. Fig. 3 (top right) shows the autocorrelation of the *in situ* SM (gray) and the estimated autocorrelations obtained for collocated SMOS SM with the LS, LASSO, and *naïve* methods. The effective temporal frequency of SMOS was used to filter the ground measurements and estimate the corresponding autocorrelation, but not noticeable differences were found with respect to the estimate with uniform sampling. The estimated autocorrelations curves of SMOS and *in situ* SM at the REMEDHUS site are similar, with a narrower main lobe given by the LS and *naïve* estimators, which causes some slight bias on the crossing with $1/e$ (e-folding time). All the methods provide a slightly reduced estimation of the autocorrelation main lobe and of the e-folding time with respect to *in situ*, especially LS and *naïve* estimators. Comparisons of the gold standard (*in situ* network average) with the estimated autocorrelation for ASMR2, ASCAT, and SMOS SM collocated time series with the LS and LASSO methods are shown in Fig. 3 (bottom panel). It can be seen that, for the two methods, the three satellite time series provide a qualitatively similar autocorrelation profile, with a narrower main lobe for ASCAT and a larger main lobe for AMSR2. A wider estimation of the autocorrelation main lobe is obtained for all satellites when using LASSO than when using LS, though all of them tend to provide narrower main lobes, and in turn to reduce the e-folding time estimates with respect to *in situ* estimates. Our results are in line with previous studies reporting a faster soil drying in microwave-based SM data than in ground observations [57], [58]. They attribute this difference to a shallower sensing depth in satellites, especially after rain events, since *in situ* probes are centered at 5-cm underground and hence do not sense the top soil. Rain events create positive vertical moisture gradients, with the near-surface soil being wetter than deeper soils shortly after rainfall and also drying out more quickly due to evaporation and vertical redistribution. Also, note that the sensing depth of microwaves is shallower for wet soils, further accentuating the combined effects of vertical SM gradients and different sensing depths [28].

Comparing satellite estimates, the fact that ASCAT displays lower persistence than SMOS (lower autocorrelation values for most time lags) is in agreement with the theory: a shallower layer of the soil contributes to surface emissivity at C-band than at L-band (1 cm versus 5 cm) therefore leading to a lower memory. However, AMSR2 displays larger persistence that the other two sensors, which seems counter-intuitive. Although the distinct properties shown by AMSR2 and ASCAT can (in part) be due to their operating mode (passive versus active sensing), the larger persistence of AMSR2 versus SMOS cannot be explained with the theoretical sensing depth of microwave frequencies. A possible interpretation of this result is that the autocorrelation estimate of AMSR2 is impacted by its lower sampling frequency (see Fig. 2). Missing short-term variability could lead to overestimation of the memory timescale [10]. Interestingly, a previous study comparing satellite and modeled

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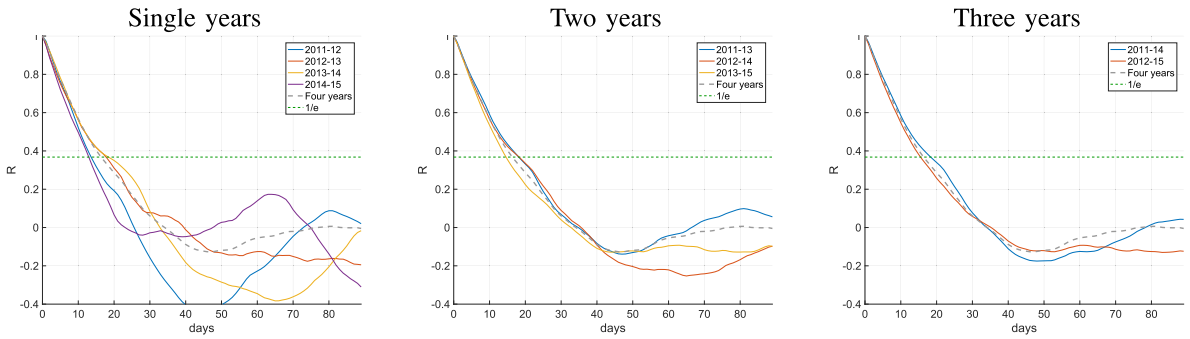


Fig. 4. Comparison between autocorrelations for REMEDHUS *in situ* SM data using the LS method and different time series length: single years (left), groups of two years (middle), and groups of three years (right). Computation using the four year study period is included in all plots as a reference and the green dashed horizontal line indicates where the autocorrelation equals $1/e$ corresponding to the e-folding time.

SM dry-downs showed that the effective sampling frequency of the satellite instruments had a greater impact than soil depth in the drydown time scale, with sparse temporal coverage missing short drydowns [59]. Despite these differences, it is remarkable that the temporal response of the three satellite data sets is very similar among them and also with respect to ground measurements.

B. Impact of Time Series Length and Missing Data

Here, we examine to what extent the measures of autocorrelation proposed to characterize SM persistence can be impacted by the time series length as well as by the possible lack of data during the winter season.

In Fig. 4, we investigate how the estimated autocorrelations vary when different combinations of years are used in its computation (1 year *versus* 2, 3, and 4 years). Results are shown for the LS method, but similar results were obtained for LASSO and *naïve* approaches. Somehow different autocorrelation curves are obtained when time series for single years are used for computation. The autocorrelation functions obtained for the combination of 2 consecutive years appear to be more similar among them, but only up to 40-day lags. The spread of autocorrelation curves is significantly reduced when combining three years. These results indicate that autocorrelation-based SM memory estimates are impacted by the use of relatively short time series and that at least 3 years of daily measurements should be used in their computation.

Fig. 5 illustrates the problem of not having data during the full year to compute the autocorrelation. In the first row, we show the autocorrelation computed using all the *in situ* data available, i.e., all year seasons included. In the second row, we show the autocorrelation computed simulating that we do not have data for the winter months (December to February), a rather typical situation for many regions where seasonal snow masks soil emissivity and hence SM retrievals cannot be attempted. As we can observe, all methods obtain similar autocorrelation curves when all data is available, with the *naïve* method leading to slightly larger e-folding times (of 16 days *versus* 11 and 12 obtained with LS and LASSO for the 4-year estimate). However, when no data is available during the winter season, the autocorrelation series greatly differ from those obtained with the full years. The most robust method is LS. As expected, the *naïve* method exhibits lower quality, and autocorrelation estimations larger than 20 days are not trustable. More surprisingly, LASSO seems to be very sensitive to the presence of long (3-months) gaps, especially when using time series of less than 3 years. In general, one can alleviate the missing data problem in the winter season in all methods by using longer time series. Also, our results show that excluding the winter months in the computation lowers the estimate of the persistence time scales in all cases. This result is in line with results reported in a previous study which conducted a similar experiment but at monthly scales [16].

C. Estimation Algorithms and Parameter Maps

The proposed LS and LASSO autocorrelation estimators are here compared temporally and spatially with the *naïve* estimator, which ignores the missing data and assumes uniform sampling. For each satellite (SMOS, AMSR2, and ASCAT), a feature map was obtained from each estimator with two parameters, namely, the τ_e time delay of the estimated autocorrelation, and the value of the autocorrelation for a fixed time delay ($\tau = 5$ days), which we denote as $R_m(5)$. Note that the former is the e-folding time parameter which is generally used to determine the lag beyond which systems are no longer autocorrelated, whereas the latter is introduced here to complete the view of the main lobe behavior in the estimated autocorrelation. For completeness, the e-folding time obtained by fitting an AR(1) model using the TAUEST methodology in [44] was also included in the comparisons.

Fig. 6 illustrates the benefits of using spatial–temporal processing for denoising the spatial distribution of temporal autocorrelations after (16). We show the results obtained for the AMSR2 satellite because it was the one in our experiments with more disagreements among methods and noise, so it represents in this setting a worse-case scenario. ASCAT and SMOS satellites resulted in slightly more reduced noise figures in general. The autocorrelations estimated with the *naïve* method [example in panel (d)] were in general highly noisy, which is very problematic for the estimation of metrics such as τ_e and $R_m(5)$. The e-folding estimates obtained with the AR(1) model corresponded to clean but extremely fast-decaying autocorrelations and extremely lower values of estimated parameters (not shown). This indicates that the SM

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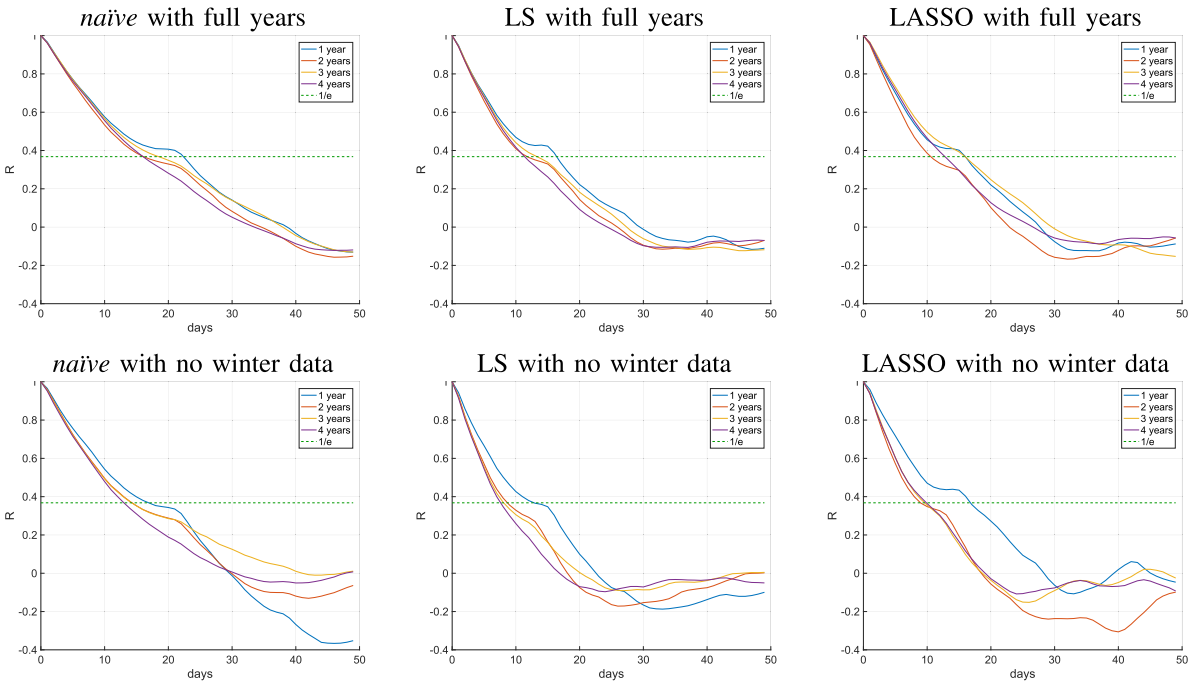


Fig. 5. Comparison between *naïve* autocorrelation method, LS and LASSO using full time series of REMEDHUS *in situ* SM data (top row) and simulating that no data is available for the winter months (December to February) (bottom row) for different time series lengths (1 to 4 years).

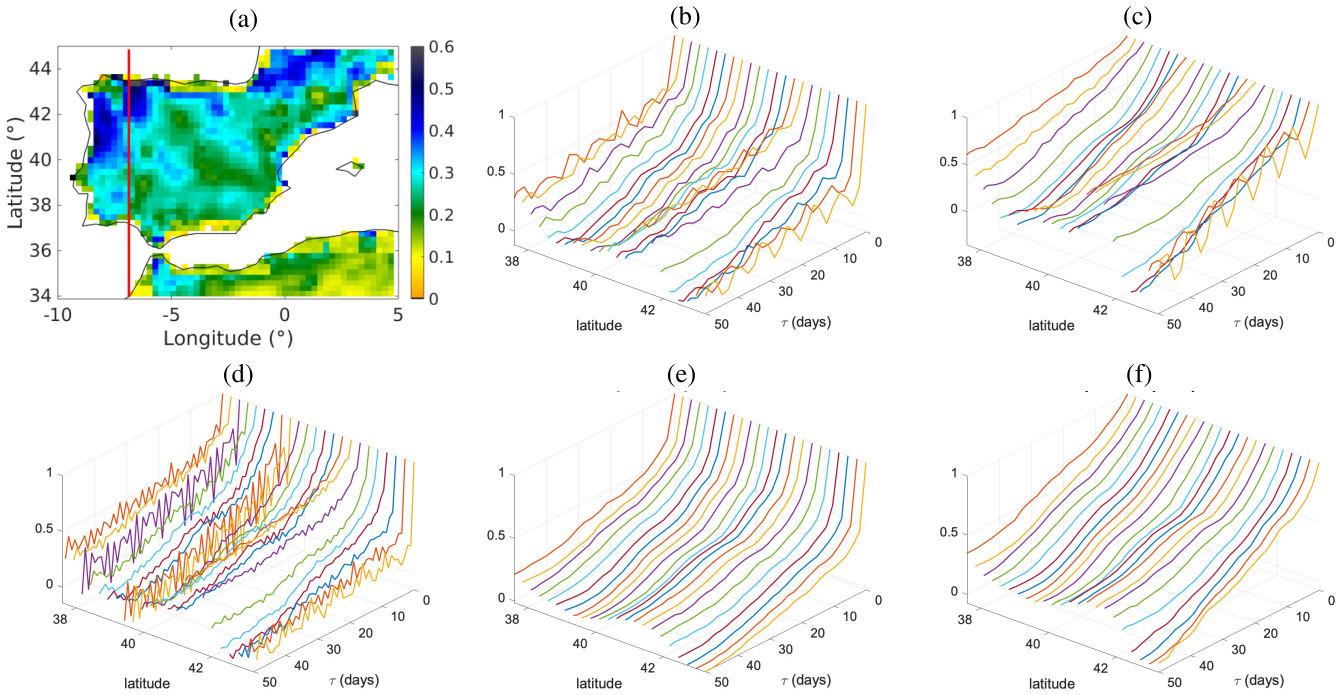


Fig. 6. Study of the impact of raw and spatial–temporal processing on the autocorrelation estimates, in AMSR2. The map of AMSR2 SM average (2012–2016) [m^3m^{-3}] indicates with a red line the chosen transect to illustrate the performance of the different methods.

time series cannot be adequately modeled by an AR(1) model, as was confirmed by simple spectral analysis (see Fig. 7).

LS and LASSO worked consistently better than TAUEST and *naïve* methods, which generally led to noisier estimates of the autocorrelation main lobe, and to lower values of persistence parameters. We noted LS and LASSO also provided noisy autocorrelation estimates on particular pixels, often due to partial errors in the estimation process of the autocorrelation. However, as shown in panels (e) and (f), these could be alleviated by spatial–temporal processing resulting in more robust estimates. For smoothing the autocorrelation in both space and time, we applied a $3\times3\times3$ block median

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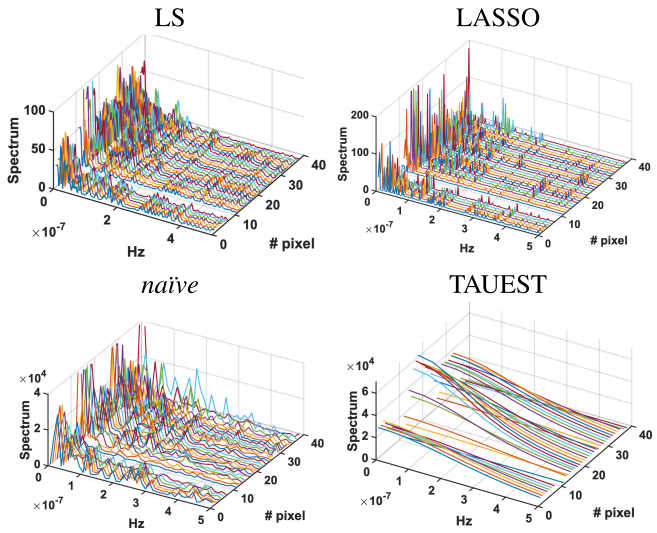


Fig. 7. Spectral analysis of the transect in Fig. 6(a), for the different persistence estimators. The LS and LASSO interpolated spectral profiles are similar between them. In the case of LS, the harmonic peaks can be seen as consistently present through pixels, whereas in LASSO these peaks are either consistent through pixels or not present, as a consequence of its L1-norm nature. The spectrum of the *naïve* estimator shows that a good part of the spatial-spectral structure is missing if we do not interpolate. The spectrum of the TAUEST estimator corresponds to the spectrum of an AR(1) process, which cannot capture with enough quality the dynamics of the time processes for any pixel.

filter, accounting for a close neighborhood in the two spatial dimensions and in the temporal dimension, hence, a matrix block with size 3 was selected as a quality trade-off between variance and bias. The median-based filter was adapted to especially obtain more accurate estimations both on the main lobe of the autocorrelation and on the coastal pixels in two ways. First, empty measurements corresponding to either deleted or to water (sea, rivers, and similar) pixels were substituted by not-a-number values. Second, the median in each block was calculated by discarding the not-a-number values, which was seen to perform better close to water and other boundaries. Finally, blocks including the unity in the zero-lag autocorrelation were first concatenated with a subblock of two consecutive ones, and then trimmed out of said ones, which avoided the over-smoothing in the close neighborhood of the zero lag.

In agreement with previous examples shown with synthetic data and over REMEDHUS (Figs. 1 and 3), the transect analyses confirm LS (LASSO) tends to give narrower (wider) autocorrelation main lobes. Note that this difference is not substantially affected by the application of the 3-D median filter.

The impact of spatial–temporal denoising can be further scrutinized on the parameter maps obtained with each method. Fig. 8 shows the parameter maps obtained with LS and LASSO methods for the AMSR2 satellite, which as previously indicated was our worst case from the point of view of noise and effective temporal frequency. The panels show a surface representation of the estimated parameter as a function of longitude and latitude. Vertical scales have been adjusted to allow visual inspection of the spatial similarity of the patterns given by each method and parameter, and the scale bottom for τ has been adjusted between 0 and 50 and 150 days (so that artifacts can be readily observed as holes or vertical bars), and for $R_m(5)$ between 0 to 1. Both parameters, τ_e and $R_m(5)$, exhibited extremely low values, in comparison with any of the other methods, when coming from TAUEST autocorrelation estimator, and both parameters tended to also have noisy and low values when using the *naïve* method (not shown).

Several conclusions can be drawn from Fig. 8, which can be extended to the results of the maps obtained for the other satellites. First, τ_e measurements are likely to provide a number of artifact measurements, mostly due to noise in the autocorrelation function leading to a strongly overestimation of its first crossing with the $1/e$ value. The LS algorithm exhibited more sensitivity to artifacts in this parameter than LASSO. This was partially alleviated by the spatial–temporal filtering, but remained present. LASSO was the most robust method with respect to artifacts in the estimation of τ_e, and improved (not totally, but significantly) when using the spatial–temporal filtered autocorrelation as a basis for the parameter calculation. Second, the spatial profiles provided by $R_m(5)$ parameters were much more robust against artifacts, and they were almost-totally suppressed when using the spatial–temporal filtered autocorrelation. Note that, despite the apparent difference between LS and LASSO estimates for this parameter, they are mostly due to an effect of a wide estimated main lobe by the LASSO algorithm. It is worth mentioning that in the other two satellites with higher temporal frequency of observations this bias in LASSO was considerably reduced, and both estimators performed similarly. In conclusion, the $R_m(5)$ parameter provides a more robust estimate or description of the SM autocorrelation function than the τ_e parameter. Also, LS tends to give narrower main lobes and it is somehow more sensitive to noise than LASSO, whereas the latter tends to give wider main lobes. Finally, the spatial–temporal denoising of the autocorrelation before retrieving any parameter map is recommended as it provides a substantial beneficial effect on its spatially concurrent patterns and robustness to artifacts.

We checked that these conclusions remained valid for the other satellites considered in the study. As an example, a complete set of parameter maps (including *naïve* and TAUEST) for the SMOS satellite are shown in Fig. 9, all of them after spatial–temporal denoising of autocorrelation tensors. In this case, the use of the *naïve* estimator yielded parameter spatial distributions closer to LS estimator, but with more artifacts, both in τ_e and in $R_m(5)$ parameters. LASSO estimators showed similar patterns when compared with LS, but with a trend to overestimate both parameters. Finally, parameters obtained from TAUEST exhibited a dramatic trend to underestimate the values with both parameter maps.

D. Soil Moisture Persistence Patterns Captured by SMOS, AMSR2 and ASCAT Satellites

The spatial patterns of persistence obtained from SMOS, AMSR2, and ASCAT satellites across the Iberian Peninsula are here analyzed to identify common features and characterize interdependencies with SM regimes. For each satellite, a map

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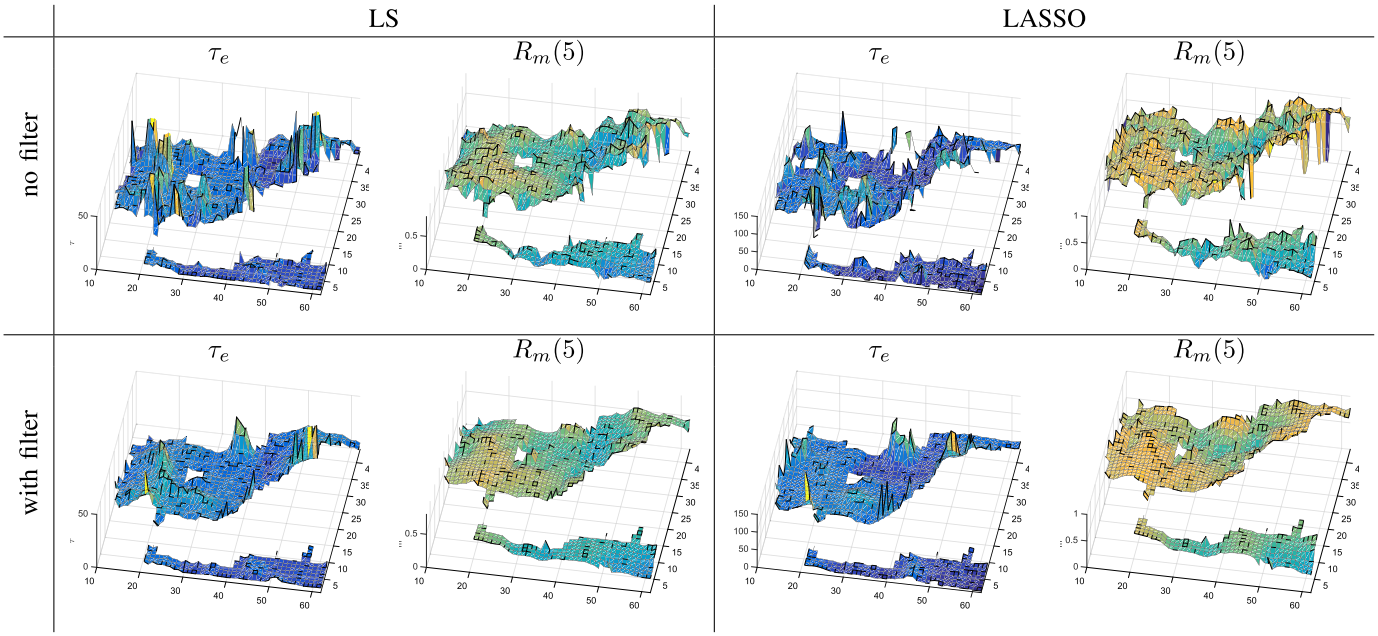


Fig. 8. Surface maps for parameters τ_e and $R_m(5)$, with LS and with LASSO, when estimated before (upper row) and after (lower row) spatial–temporal denoising of the autocorrelation tensor.

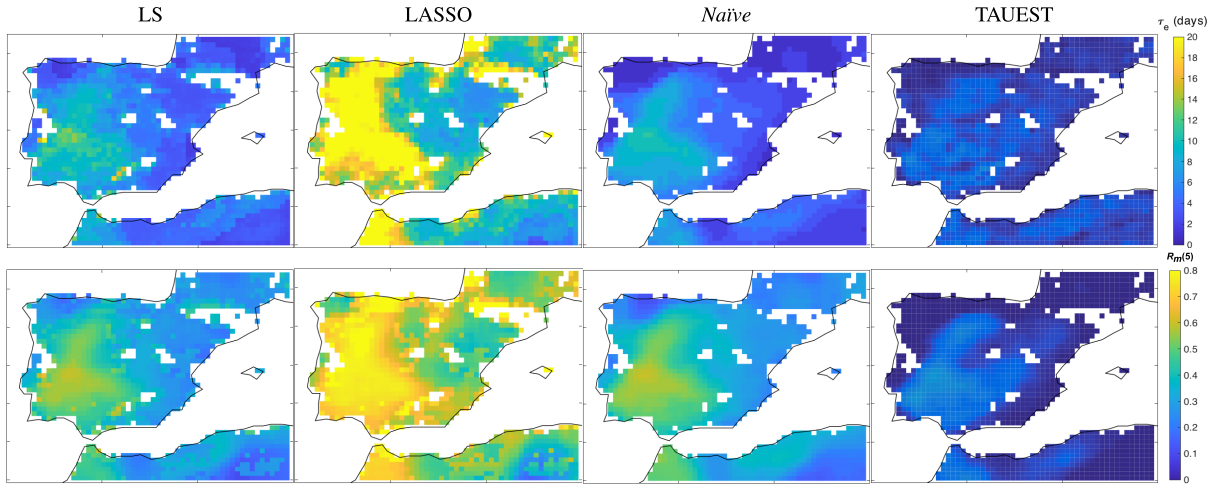


Fig. 9. Maps of τ_e (top) and $R_m(5)$ metric (bottom) obtained for the SMOS SM data with the considered methods (the proposed LS and LASSO, as well as *naïve* and TAUEST as reference), after spatial–temporal denoising of the autocorrelation tensor.

of τ_e and a map of $R_m(5)$ were obtained as measures of SM persistence. The spatial variation of these maps for each method (the proposed LS and LASSO as well as *naïve* and TAUEST for reference) are first compared in the histograms shown in Fig. 10, all of them after spatial–temporal smoothing of autocorrelation tensors. Results show: 1) the low performance of TAUEST; 2) the similarity of LS and *naïve* results, with the latter still showing a few artifacts near zero after the spatial–temporal denoising; and 3) that LASSO leads to the largest parameter estimates, due to the overestimation of the autocorrelation main lobe as discussed in the previous experiments. Interestingly, for all methods, larger persistence measures are obtained from AMSR2 than from SMOS and ASCAT satellites, which distributions are more closely aligned. This suggests theoretical differences in sensing depth between L- and C-band sensors (of 5 versus 1 cm) have a reduced impact on the persistence dynamics captured by the different satellites. The larger persistence estimates of AMSR2 could instead be explained by the fact that the AMSR2 product has a lower temporal frequency than the other two products (see Fig. 2), and hence it may be missing substantial short-term variability that the other two products are actually capturing. Despite these differences, Fig. 11 (two first rows) shows that the spatial patterns of persistence captured by the three satellites are consistent. This is remarkable, given the differences in sensing technologies and retrieval approaches implemented in each of the satellite missions. These results support the combination of data from C- and L- active and passive microwave sensors in long-term observational records as in C3S.

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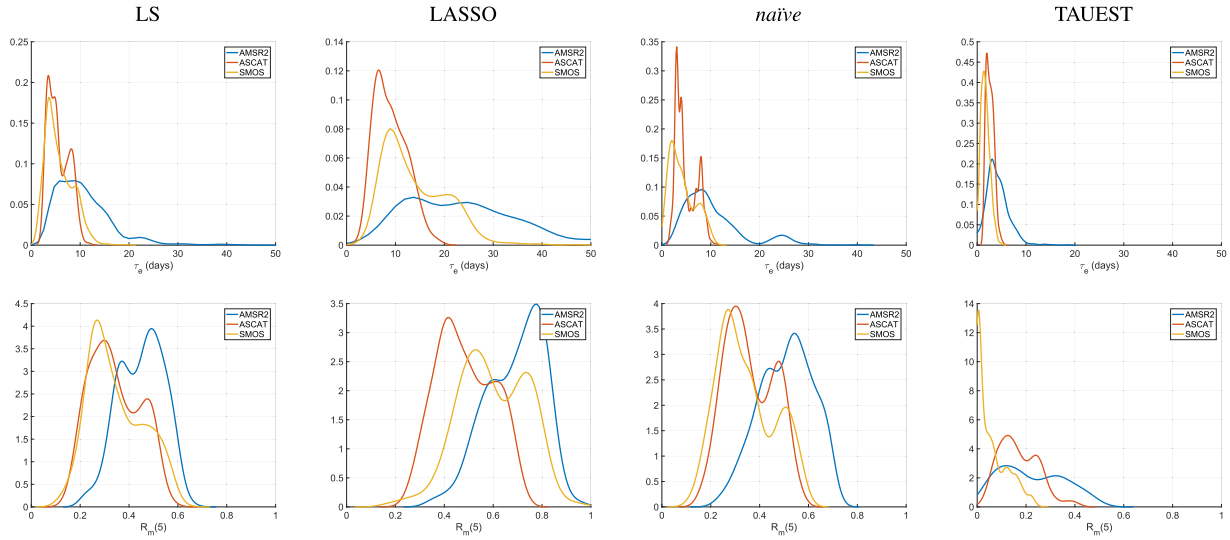


Fig. 10. Histograms with τ_e (top) and $R_m(5)$ metrics (down) for different methods and after spatial–temporal denoising of the autocorrelation tensor.

A quantitative measure of the spatial correlation of the obtained τ_e and $R_m(5)$ maps for the three satellites is provided in Fig. 12. Note the spatial changes are smoother for ASCAT and SMOS (of 0.68 and 0.72 for τ_e and $R_m(5)$, respectively), and less smooth for ASCAT and AMSR2 (of 0.29 and 0.53 for τ_e and $R_m(5)$, respectively). In general, maps obtained for $R_m(5)$ are smoother than for τ_e.

The dependence of the SM persistence patterns on the SM regime, characterized by the average SM content for the study period, is investigated in Fig. 11 (last row). Results are shown for the SMOS satellite but similar patterns of SM memory and interdependencies with SM regime were obtained for the three sensors (not shown). The analysis illustrates that common features can be identified in the relationship between SM memory, expressed here by τ_e and $R_m(5)$ metrics, and SM regime. SM memory is lowest in dry areas, and also decreases at high SM values, resulting in a bell-shaped relationship, with the highest SM memory values at intermediate soil wetness. Still, note the spread is large and the differences small between the different grouped-pixels and therefore this result should be taken with caution, and confirmed with further studies, preferably conducted at global or continental scales. This bell-shaped relationship of SM memory with SM regime was also reported in a previous study conducted at monthly scales using eight different atmospheric general circulation models [21]. The analysis of the processes controlling SM memory in the models demonstrated that it was mostly controlled by two effects, namely, evaporation sensitivity to SM, which increases with decreasing SM content, and runoff sensitivity to SM, which increases with increasing SM content. Thus, larger values of SM memory were observed in regions of medium SM content, where both effects are small. Our results, although at a regional scale, support this hypothesis with observational data. Still, note other studies report shorter drydown scales (lower memory) for drier soils (e.g., [10], [59]). Importantly, other factors such as vegetation type or soil texture, apart from meteorological forcing, could also partially explain the obtained surface SM persistence patterns and should be a subject of further research.

V. Discussion and Conclusion

Memory and persistence are ubiquitous concepts in many Earth observation problems. Characterization of SM persistence has important implications in hydrology, vegetation, and water modeling and understanding. This work addressed the issue of characterizing and estimating soil memory and persistence from microwave satellite data.

The problem is very challenging statistically because of the uneven sampling, short and long-term persistence, and intertwined dynamic processes. We compared several strategies and showed results in both simulated and real observational SM products from different sensors. In particular, we have paid attention to the bottleneck of persistence estimation, namely, the appropriate estimation of the autocorrelation function in nonuniformly sampled SM time series. Several loss functions and scores summarizing the SM autocorrelation function have been proposed. We illustrated their performance using ground-based SM measurements from a high-quality network, along with remotely sensed SM time series acquired by L-band passive (SMOS), C-band passive (AMSR2), and C-band active (ASCAT) microwave sensors. In general, accounting for the nonuniform nature of the time series using improved autocorrelation estimators has provided consistent spatial descriptions of SM persistence. Let us discuss further some relevant aspects of the study.

On the impact of anomalies and signal preprocessing. The proposed metrics to estimate persistence are dependent on the estimated mean SM state used to calculate the anomalies, which can be uncertain (especially for short temporal records). For this study, we showed that at least 3 years of daily observations were required to obtain improved estimates of the autocorrelation function. Including more sophisticated preprocessing steps, e.g., to better account for the seasonal cycle, could certainly have an effect on the methods performance in general and on LASSO in particular. Note that the autocorrelation function and derived persistence metrics ignore the sign of the anomaly, which can yield meaningful information as suggested in [10].

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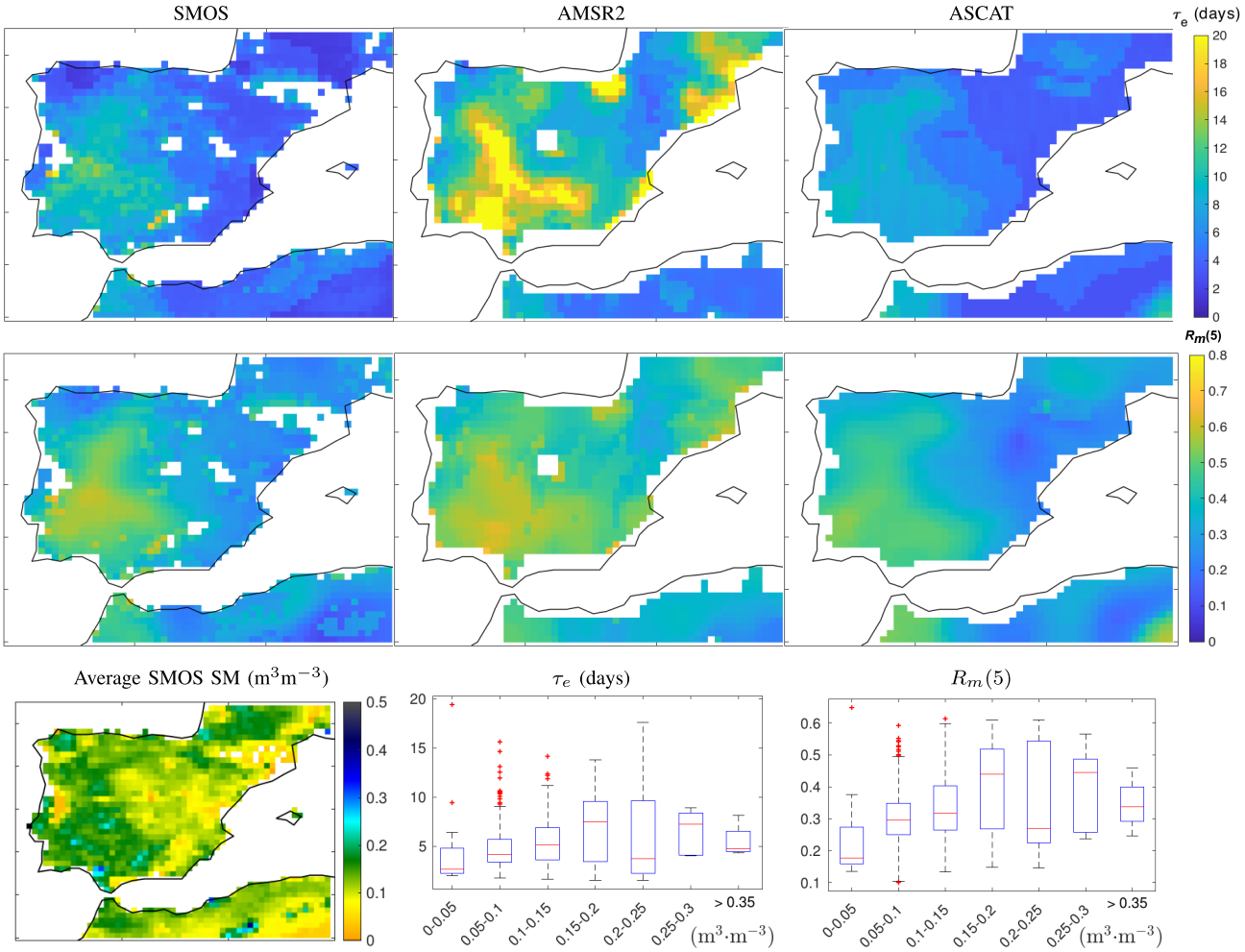


Fig. 11. Maps of estimated autocorrelation metrics τ_e (top row) and $R_m(5)$ (second row) obtained for SMOS, AMSR2 and ASCAT SM data, using the LS estimator after spatial–temporal denoising of the autocorrelation sensor. Third row: Relation of the τ_e and $R_m(5)$ maps obtained for SMOS and the mean map of SMOS SM average (2012–2016) [m^3m^{-3}] (left panel). Boxplots show the median (red line) values, and the boxes (blue) mark the extent of the 25th and 75th percentiles. The whiskers extend to the maximum and minimum values.

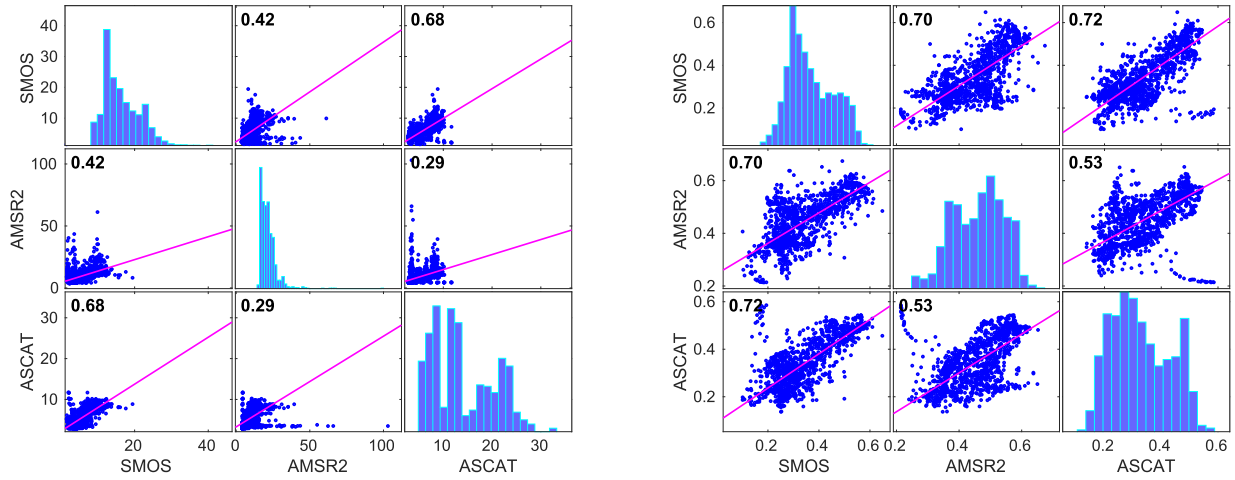


Fig. 12. Correlation matrix of the persistence patterns captured by SMOS, AMSR2 and ASCAT satellites, expressed as the proposed autocorrelation metrics τ_e (left) and $R_m(5)$ (right). Parameter maps are estimated using the LS estimator, after spatial–temporal denoising of the autocorrelation sensor.

On the spatial structure of autocorrelation and persistence. The autocorrelation function was estimated for each time series individually, which yielded particularly noisy estimates in the spatial domain. The analysis of transects allowed us to thoroughly assess the consistency of the methods and the spatial–temporal characteristics of the autocorrela-

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tion estimators and derived metrics. This visualization also allowed us to introduce simple spatial processing stages, such as the adapted median filtering of the temporal autocorrelations, which largely improved results in all cases. As far as smooth spatial changes are present, it is very likely that local spatial–temporal autocorrelation estimates could yield better estimators. Future work will consider extending the estimation of spatial–temporal autocorrelation tensors. Other estimation methods, such as kernel methods (i.e., support vector machines or Gaussian processes) based on the use of autocorrelation kernels or autocorrelation covariance matrices, could be explored as an alternative framework for this application to better cope with possible nonlinearities and nonstationarities.

On the Autocorrelation main lobe and larger lags. Several estimation methods can give acceptable results for the main lobe of the temporal autocorrelation, even though the two better performing methods (LS and LASSO) may exhibit some estimation bias. Although in this study we focus on the main temporal lobe of the autocorrelation function, we noted that estimates at lags larger than 20 days exhibit strong variability, consistently with the spectral profile observed in the time series (not shown). Parameters derived from the main lobe of the autocorrelation function can constitute a valuable score to describe signal persistence properties. The conventional e-folding decay factor, however, should be either checked for adequate estimation or estimated with more accurate methods that the generally used *naïve* or TAUEST in case of nonuniform sampling. Otherwise, we showed it could lead to artificial estimates. In particular, the suggested parameter $R_m(5)$ was introduced as an alternative and more robust descriptor, in the sense of its moderate insensitivity to outliers and atypical samples. Metrics like the e-folding time and $R_m(5)$ try to summarize the autocorrelation decay on its main lobe, and as such, they will be strongly redundant, and other parameters summarizing this property will be also informative and redundant. The use of the autocorrelation for a fixed time lag has previously been proposed in the literature as a persistence metric [8], [21]. The use of the slope, for instance, could be also informative, assumed that we choose an initial and final lag for its estimation and establish a linear parametrization that can reasonably fit a nonlinear decay of the main lobe lags, and then that the chosen approximation holds in a high number of pixels. We chose to scrutinize here the $R_m(5)$ coefficient, aiming to be somewhere in the middle of the zero lag and of the first zero crossing, but other lags could also be useful, such as $R_m(4)$ for being closer to the zero lag. In our data, a variety of vegetation and soil characteristics is present, and both possibilities performed well. Note that we do not stick here to the proposal of using lag 5, but rather on using an intermediate autocorrelation lag between the zero lag and the first zero crossing as a more straightforward index than the e-folding time, according to our experiments. The extension of this study to other regions will allow us to establish the metrics that can be globally adequate. Yet, other metrics could be used to capture the SM dynamics too, whose definition is beyond the scope of the present work. It goes without saying that new descriptors and parameters will need to consider the signal characteristics explicitly. For instance, while sparsity might seem a sensible criterion, we hypothesize that the LASSO algorithm could be suboptimal as the spectrum is not necessarily sparse but rather wideband.

Seasonal aspects of SM persistence. Our results suggest that the estimated metrics have seasonal variation, with maximum values in winter. However, the limited number of daily samples within a season does not allow us to obtain improved estimates of temporal autocorrelation per season and to investigate further the seasonal dependence. Other techniques such as the ones focused on the estimation of the dry-down time scale [58], [59] or on detrended fluctuation analysis [60] may be more adequate to characterize persistence at short time scales.

SM persistence patterns captured by microwave satellites. Despite differences in sensing technologies and retrieval approaches, our results show that the SM products from the three microwave satellites present rather similar patterns of SM persistence in semiarid regions such as the Iberian Peninsula. Larger persistence estimates were obtained for AMSR2 than with SMOS and ASCAT, which could (at least partially) be explained by the fact that the AMSR2 product has a lower temporal frequency than the other two sensors (only one overpass is integrated into the product), and hence may be missing short-term variability that the other two sensors are capturing. Knowledge of the SM persistence captured across microwave frequencies and the impact of effective temporal sampling is particularly relevant for the planned Copernicus Imaging Microwave Radiometer (CIMR) mission [61], which will provide simultaneous multifrequency observations (including L- and C-bands) with a high temporal sampling (< 1 day). Compared to *in situ* data, our results showed faster soil drying in microwave-based SM data, which could be explained by combined effects of vertical SM gradients (the top surface being wetter after a rain event) and their different sensing depths (probes placed at a 5 cm depth). Our analyses suggest that SM persistence is higher in areas with intermediate values of SM, where run-off and evaporation water losses are small. Investigation of interdependencies with other factors such as soil properties and vegetation type should be subject of further research, preferably conducted at a global scale to capture maximum variability. Note the similar autocorrelation functions and spatial–temporal patterns of persistence obtained with the three microwave sensors support their combination in long-term observational records to be used for climate studies. Yet, the presented results should be confirmed in other regions under different climate and vegetation conditions.

On the contributions of the present work. The main motivation and our endeavor here was to aboard the estimation of the autocorrelation function, and from there to explore some reasonable metrics to quantify persistence in satellite SM time series. This has been approached from many different perspectives and with many strategies, but we observed that the impact of nonuniform sampling was larger than expected, and even overlooked to a large extent by the community. This is why we decided to take a step back and approach the problem from a purely statistical point of view without prior beliefs. The techniques deployed in our study are quite standard in signal processing and statistics, and their choice was not incidental.

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Our aim has been to get most of the information from the signal with standard tools in such a challenging scenario of unevenly distributed time series. Can one assume that L_2 regularization (hence a Gaussian likelihood noise model) is plausible enough, or is an L_1-norm implemented in LASSO (and hence a sparse-promoting driving mechanism of memory and persistence) better suited? Other scientific and technical questions appeared in our pathway too. The fact that many publications do not even justify the chosen method to estimate the autocorrelation function (which has a dramatic impact as evidenced in our work) or the metrics derived thereof, and both in such complex scenarios and regimes, encouraged us to take this challenge.

An accurate and solid estimation of autocorrelation metrics may serve as a framework for explaining inter-model differences in simulated SM memory and confront them with observation-based estimates. Investigating the differences between observed and modeled persistence estimates and their possible relation to the mechanistic models implemented in the considered land surface model or reanalysis product, for example, is a matter of future research. Such exercise could be helpful to improve parameterization in climate modeling, especially in those models where complex dynamics and memory effects are poorly represented or even not considered at all. Processes like the impact of deeper layers on the upper layers in times of water stress [16] could be better resolved with a proper SM memory characterization. Learning appropriate parameterizations, dynamics, and persistence parameters directly from observations may reduce the sources of uncertainties in current models, eventually leading to a breakthrough in climate modeling.

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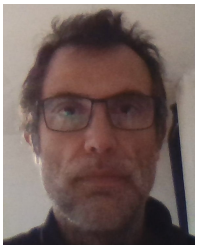
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